Unseen gene perturbation in Replogle22 K562

a WMSE

Technical duplicate — $P = 4.7e-65$, $t = -17.9$
Interpolated duplicate — $P = 1.2e-52$, $t = -16.0$
scGPT — $P = 1.4e-24$, $t = -10.7$
PRESAGE — $P = 2.9e-74$, $t = -19.3$
fMLP-GenePT — $P = 1.5e-44$, $t = -14.6$
fMLP-UCE (Δ+protein) — $P = 9.9e-34$, $t = -12.6$
fMLP-ESM2 — $P = 3.5e-35$, $t = -12.9$
fMLP-tGPT — $P = 6.9e-23$, $t = -10.3$
fMLP-scGPT — $P = 1.1e-14$, $t = -8.2$
fMLP-Geneformer — $P = 4.3e-12$, $t = -7.4$
GeneCompass (n=1660) — $P = 2.3e-08$, $t = -6.1$
Linear — $P = 1.5e-20$, $t = -9.7$
fMLP-Nicheformer — $P = 1.3e-08$, $t = -6.2$
scFoundation (n=1801) — $P = 1.8e-05$, $t = -4.9$
fMLP-scBERT — $P = 0.0183$, $t = -3.3$
GEARS — $P = 0.218$, $t = -2.5$
fMLP-scratch learned — $P = 2.3e-10$, $t = 6.8$
fMLP-scratch random — $P = 6.3e-09$, $t = 6.3$
Control mean — $P = 2.3e-46$, $t = 14.9$

−0.03 −0.02 −0.01 0.00 0.01
Δ metric (prediction − dataset mean) (↓)

b WR^2 (Δctrl)

Technical duplicate — $P < 10^{-300}$, $t = 47.6$
Interpolated duplicate — $P = 2.5e-284$, $t = 43.8$
PRESAGE — $P = 3.3e-179$, $t = 32.2$
scGPT — $P = 1.5e-77$, $t = 19.8$
fMLP-GenePT — $P = 1.2e-91$, $t = 21.7$
fMLP-ESM2 — $P = 1.5e-55$, $t = 16.5$
fMLP-UCE (Δ+protein) — $P = 1.0e-21$, $t = 10.0$
fMLP-tGPT — $P = 3.7e-28$, $t = 11.5$
fMLP-Nicheformer — $P = 2.0e-20$, $t = 9.7$
fMLP-scGPT — $P = 1.6e-10$, $t = 6.9$
fMLP-Geneformer — $P = 4.6e-09$, $t = 6.4$
Control mean — $P = 2.2e-08$, $t = 6.1$
Linear — $P = 2.3e-24$, $t = 10.6$
scFoundation (n=1801) — $P = 0.0172$, $t = 3.3$
fMLP-scBERT — $P = 5.4e-15$, $t = -8.3$
fMLP-scratch learned — $P = 1.4e-43$, $t = -14.4$
GEARS — $P = 5.4e-56$, $t = -16.5$
fMLP-scratch random — $P = 2.2e-92$, $t = -21.8$

0.0 0.2 0.4 0.6
Δ metric (prediction − dataset mean) (↑)

c Pearson (Δpert)
raw score vs 0; mean predictor undefined

Technical duplicate — $P < 10^{-300}$, $t = 62.5$
PRESAGE — $P < 10^{-300}$, $t = 47.6$
Interpolated duplicate — $P < 10^{-300}$, $t = 51.7$
fMLP-GenePT — $P = 1.9e-222$, $t = 37.0$
fMLP-ESM2 — $P = 1.3e-170$, $t = 31.3$
fMLP-UCE (Δ+protein) — $P = 7.0e-141$, $t = 27.8$
fMLP-tGPT — $P = 9.5e-111$, $t = 24.2$
fMLP-Nicheformer — $P = 2.4e-91$, $t = 21.7$
fMLP-scGPT — $P = 1.6e-104$, $t = 23.4$
fMLP-Geneformer — $P = 1.8e-90$, $t = 21.5$
scGPT — $P = 6.4e-116$, $t = 24.8$
Linear — $P = 2.3e-42$, $t = 14.2$
scFoundation (n=1801) — $P = 8.0e-44$, $t = 14.5$
Control mean — $P = 7.6e-12$, $t = 7.3$
fMLP-scBERT — $P = 5.9e-10$, $t = 6.7$
GEARS — $P = 2.3e-06$, $t = 5.3$
fMLP-scratch learned — $P = 1$, $t = 1.7$
fMLP-scratch random — $P = 2.5e-04$, $t = -4.4$

0.0 0.1 0.2 0.3 0.4 0.5
Pearson(Δpert), mean over perturbations (↑)

d MSE

Interpolated duplicate — $P = 1.0e-60$, $t = -17.3$
PRESAGE — $P = 1.7e-50$, $t = -15.6$
fMLP-GenePT — $P = 8.9e-05$, $t = -4.6$
fMLP-ESM2 — $P = 0.00149$, $t = -3.9$
Linear — $P = 1.8e-15$, $t = -8.4$
fMLP-tGPT — $P = 1$, $t = -0.1$
fMLP-UCE (Δ+protein) — $P = 1$, $t = 1.8$
fMLP-Nicheformer — $P = 0.114$, $t = 2.7$
scGPT — $P = 6.6e-12$, $t = 7.3$
fMLP-scGPT — $P = 1.5e-04$, $t = 4.5$
scFoundation (n=1801) — $P = 1.2e-08$, $t = 6.2$
fMLP-Geneformer — $P = 4.3e-11$, $t = 7.1$
Control mean — $P = 1.7e-80$, $t = 20.2$
fMLP-scBERT — $P = 2.2e-37$, $t = 13.3$
fMLP-scratch learned — $P = 3.0e-165$, $t = 30.7$
Technical duplicate — $P = 8.4e-27$, $t = 11.2$
GEARS — $P = 1.6e-171$, $t = 31.4$
fMLP-scratch random — $P = 7.6e-192$, $t = 33.7$

−0.0010 −0.0005 0.0000 0.0005 0.0010 0.0015 0.0020
Δ metric (prediction − dataset mean) (↓)

e Pearson (Δctrl)
Control mean omitted (score undefined)

Technical duplicate — $P = 4.8e-147$, $t = 28.6$
Interpolated duplicate — $P = 3.2e-176$, $t = 31.9$
PRESAGE — $P = 8.3e-100$, $t = 22.8$
Linear — $P = 1.4e-34$, $t = 12.8$
fMLP-ESM2 — $P = 0.197$, $t = 2.5$
fMLP-UCE (Δ+protein) — $P = 1$, $t = 0.8$
fMLP-GenePT — $P = 1$, $t = -0.9$
scGPT — $P = 1.3e-16$, $t = -8.7$
fMLP-tGPT — $P = 3.0e-07$, $t = -5.7$
fMLP-scGPT — $P = 3.1e-17$, $t = -8.9$
fMLP-Nicheformer — $P = 1.5e-20$, $t = -9.7$
scFoundation (n=1801) — $P = 1.6e-39$, $t = -13.7$
GEARS — $P = 1.1e-121$, $t = -25.5$
fMLP-scBERT — $P = 2.8e-59$, $t = -17.0$
fMLP-Geneformer — $P = 1.1e-40$, $t = -13.9$
fMLP-scratch random — $P = 1.0e-131$, $t = -26.7$
fMLP-scratch learned — $P = 9.2e-170$, $t = -31.2$

−0.10 −0.05 0.00 0.05 0.10 0.15 0.20
Δ metric (prediction − dataset mean) (↑)

f NIR

Technical duplicate — $P < 10^{-300}$, $t = 49.2$
PRESAGE — $P = 7.8e-141$, $t = 27.8$
Interpolated duplicate — $P = 2.4e-91$, $t = 21.7$
fMLP-GenePT — $P = 1.2e-94$, $t = 22.1$
fMLP-ESM2 — $P = 2.8e-71$, $t = 18.9$
fMLP-UCE (Δ+protein) — $P = 2.4e-70$, $t = 18.7$
fMLP-tGPT — $P = 1.1e-47$, $t = 15.2$
fMLP-Geneformer — $P = 2.6e-44$, $t = 14.6$
fMLP-scGPT — $P = 2.4e-44$, $t = 14.6$
fMLP-Nicheformer — $P = 2.6e-36$, $t = 13.1$
scFoundation (n=1801) — $P = 3.0e-23$, $t = 10.4$
scGPT — $P = 3.4e-37$, $t = 13.3$
fMLP-scBERT — $P = 2.2e-11$, $t = 7.2$
GEARS — $P = 1.6e-08$, $t = 6.2$
Linear — $P = 3.4e-20$, $t = 9.6$
fMLP-scratch learned — $P = 1$, $t = -0.1$
fMLP-scratch random — $P = 1$, $t = -0.3$
Control mean — $P = 5.6e-45$, $t = -14.7$

0.0 0.1 0.2 0.3 0.4 0.5
Δ metric (prediction − dataset mean) (↑)